

James Zhu

james.zhu@laas.fr • jameswzhu.github.io

Education

PhD, Mechanical Engineering

Carnegie Mellon University

September 2024

Thesis: Navigating a Complex World: Improving Robot Outcomes Through Social, Regulatory, and Control Theoretic Approaches

MS, Mechanical Engineering

Carnegie Mellon University

December 2022

BE, Mechanical Engineering and Mathematics

Vanderbilt University

May 2020

Cum Laude

Work Experience

Postdoctoral Researcher

French National Center for Scientific Research

Supervisors: Thierry Simeon & Marco Cagnetti

Toulouse, France

January 2025 – Current

Graduate Research Assistant

Carnegie Mellon University

Advisor: Aaron Johnson

Pittsburgh, Pennsylvania

May 2020 – September 2024

Robots, Ethics, & Society Research Intern

The Robotics & AI Institute

Supervisor: Kate Darling

Boston, Massachusetts

January 2024 – June 2024

Undergraduate Research Assistant

Vanderbilt University

Advisors: Nilanjan Sarkar & Robert Webster

Nashville, Tennessee

September 2017 – May 2020

High Contrast Imaging Intern

Jet Propulsion Laboratory

Supervisor: Stuart Shaklan

Pasadena, California

May 2019 – July 2019

Engineering and Operations Intern

AT&T

Atlanta, Georgia

May 2018 – August 2018

Teaching

Teaching Consultant Fellow: CMU Eberly Center for Teaching Excellence

September 2022 – December 2023

Advisory Board Member: CMU Teaching & Learning Summit

September 2023

Teaching Assistant: Dynamics

Spring 2022 and Spring 2023

Teaching Assistant: Intro to Robotics

Spring 2019

Teaching Assistant: Probability and Statistical Inference

Spring 2019

Certifications & Training

Summer Institutes in Computational Social Science Program

June 2025

Inclusive STEM Teaching Certificate

April 2023

Tech Stewardship Practice Program Certificate

December 2022

Languages

English: Native

French: Intermediate (B1)

Mandarin Chinese: Heritage

Publications

Journal Papers

Asynchronous Distributed Multi-Robot Motion Planning Under Imperfect Communication

*Ardalan Tajbakhsh, Augustinos Saravanos, **James Zhu**, Evangelos A Theodorou, Lorenz T Biegler, and Aaron M Johnson*

Under Review: IEEE Robotics and Automation Letters

Robust Sensitivity-Aware Chance-Constrained MPC for Efficient Handling of Multiple Uncertainty Sources

***James Zhu**, Thierry Simeon, and Marco Cagnetti*

2025 IEEE Robotics and Automation Letters

Hybrid Iterative Linear Quadratic Estimation: Optimal Estimation for Hybrid Systems

*J Joe Payne, **James Zhu**, Nathan J Kong, and Aaron M Johnson*

2025 IEEE Robotics and Automation Letters

Saltation Matrices: The Essential Tool for Linearizing Hybrid Dynamical Systems

*Nathan J Kong, J Joe Payne, **James Zhu**, and Aaron M Johnson*

2024 Proceedings of the IEEE

Conference Papers

On the Parametric Sensitivity of Hybrid Systems and its Application to Robust Trajectory Optimization

Tommaso Belvedere, **James Zhu***, Marco Cagnetti, and Paolo Robuffo Giordano*

2026 IEEE International Conference on Robotics and Automation

Convergent iLQR for Safe Trajectory Planning and Control of Legged Robots

***James Zhu**, J Joe Payne, and Aaron M Johnson*

2024 IEEE International Conference on Robotics and Automation

Grounding Robot Navigation in Self-Defense Law

***James Zhu**, Anoushka Shrivastava, and Aaron M Johnson*

2023 IEEE International Conference on Robot and Human Interactive Communication

Hybrid Event Shaping to Stabilize Periodic Hybrid Orbits

***James Zhu**, Nathan J Kong, George Council, and Aaron M Johnson*

2022 IEEE International Conference on Robotics and Automation

Design and System Validation of Rasse: A Novel Active Social Assistive Robot with a User Interface for Elderly with Dementia

*Zhaobo K Zheng, **James Zhu**, Jing Fan, and Nilanjan Sarkar*

2018 IEEE International Symposium on Robot and Human Interactive Communication

In Preparation

Robust Learning-Based Robot Control Via Sensitivity-Aware Observation Augmentation

***James Zhu**, Thierry Simeon, and Marco Cagnetti*

In Preparation

Challenges and Opportunities for Autonomous Delivery Robots in Charitable Food Systems

Swapna Joshi, **James Zhu***, Pedro Reynolds-Cuellar and Kate Darling*

In Preparation

Drivers of Technology and Infrastructure Design: Reflections from a Long-Term Public Robot Implementation

***James Zhu** and Anastasia K Ostrowski*

In Preparation

Duty Factor Predicts Safe Quadrupedal Robot Locomotion

***James Zhu**, David Ologan, Selvin Garcia Gonzalez, Ardan Tajbakhsh, and Aaron M Johnson*

In Preparation

Beyond Single Outcomes: Scalable Multi-Modal Motion Planning in Dynamic Multi-Agent Environments

*Ardalan Tajbakhsh, Siddharth H Nair, David Ologan, **James Zhu**, Amey Sheth, Lorenz T Biegler, and Aaron M Johnson*

In Preparation

Other Papers

Risk-Aware Active Perception and Control in Sensing-Constrained Environments

James Zhu, Thierry Simeon, and Marco Cagnetti

2026 ICRA Workshop on Learning and Representations for Active Perception in Manipulation

Double-Anonymous Review for Robotics

Justin K Yim, Paul Nadan, James Zhu, Alexandra Stutt, J Joe Payne, Catherine Pavlov, Aaron M Johnson

2022 Technical Report: arXiv:2406.10059

By Air or by Land: How Locomotion Methods Dictate Drone Ethics

James Zhu, Thierry Simeon, and Marco Cagnetti

2022 ICRA Workshop on Addressing Ethical and Technical Challenges in the Development, Use and Governance of Lethal Autonomous Weapons Systems

Conferences, Workshops, & Seminars

Presentations

Embracing the Unknown: Robust Robotics in Uncertain Environments

LIRMM Interactive Digital Humans Seminar

December 2025

Invited Talk

Embracing the Unknown: Robust Robotics in Uncertain Environments

Loria Human-Centered Robotics Seminar

November 2025

Invited Talk

Robust RL Navigation Via Sensitivity-Aware Observation Augmentation

IROS Workshop on The Art of Robustness: Surviving Failures in Robotics

October 2025

Poster Presentation

What Can a PhD Student Learn About Robots That an AI Can't?

LAAS-CNRS Robotics and Interactions Seminar

February 2025

Invited Talk

Improving Equity in Robot Deployment

University of Tennessee Center for Biomedical Informatics Seminar

September 2024

Invited Talk

Designing Safe Quadrupedal Gaits

ICRA Advancements in Trajectory Optimization and MPC for Legged Systems Workshop

May 2024

Poster Presentation

Navigating a Complex World: How to Bring Robots Into the Wild

New York University Dynamical Systems Lab Seminar

April 2024

Invited Talk

An Engineering Perspective on Legislation Challenges for Autonomous Delivery Robots

Georgia Tech Solving for X Workshop

February 2024

Invited Talk

Grounding Robot Navigation in Self-Defense Law

WeRobot

October 2023

Poster Presentation

Convergent Planning and Control of Legged Robots

IEEE RAS TC on Model-Based Optimization for Robotics Poster Session

July 2023

Poster Presentation

Convergent Planning and Control of Legged Robots

Carnegie Mellon Mechanical Engineering PhD Symposium

March 2023

Poster Presentation

Convergent iLQR for Underactuated Hybrid Dynamical Systems

RSS Risk Aware Decision Making Workshop

June 2022

Lightning Talk

Hybrid Event Shaping to Generate Stable Robotic Gaits

Carnegie Mellon Locomotion Seminar

March 2022

Invited Talk

Organization

Georgia Tech Solving for X Workshop

2024

Session Organizer

Editorial

IEEE International Conference on Intelligent Robots and Systems
Associate Editor

2025 & 2026

Additional Conferences

Ethical and Legal Dilemmas of Autonomous Weapons in War and National Security Conference
Invited Discussant

April 2024

WeRobot
Participant

September 2022

Leadership, Honors, and Additional Experience

Ordained Minister: American Marriage Ministries

February 2024

Co-Founder & Chief Flavor Architect: Cup of Wontons

February 2023 – December 2023

Featured in 2024 CMU Engineering Magazine article

Equity Researcher: Equitable and Just Greater Pittsburgh

December 2022 – December 2023

Organizer: Gwen's Girls Robotics Outreach Program

Fall 2020 – Fall 2022

Featured in 2023 CMU Engineering Magazine article

Co-Chair: Mechanical Engineering DEI Outreach Subcommittee

January 2021 – May 2022

Carolyn Commer Graduate Student Involvement Award

May 2021

Student Mentor: Carnegie Mellon Tartan Scholars Program

August 2020 – May 2021

Schiff Family Scholarship

2018–2020

Attraction Operations Specialist: EnterTRAINment Junction

June 2017 – January 2018

Adult Literacy Program Volunteer: Nashville Public Library

January 2017 – May 2017

Students Advised

José Otavio Brochado Colombini

January 2026 - Current

PhD in Robotics, University of Toulouse

Riccardo Fuser

January 2025 - Current

PhD in Robotics, University of Toulouse

Jonas Soueidan

January 2025 - Current

PhD in Robotics, University of Toulouse

Ahmed Yousef

June 2025 - April 2026

MS in Robotics Engineering, University of Genova

Joshua Ramos

February 2022 - September 2024

BS in Electrical Engineering, CMU

Selvin Garcia Gonzalez

February 2024 - July 2024

MS in Mechanical Engineering, CMU

Karla Soto Cuevas

September 2023 - December 2023

MS in Mechanical Engineering, CMU

Sasha Kroman

September 2023 - December 2023

BS in Mechanical Engineering, CMU

Nikhil Chinnalapatti Gopinath

June 2023 - December 2023

MS in Mechanical Engineering, CMU

Anoushka Srivastava

January 2023 - May 2023

BS in Artificial Intelligence, CMU